

Minimum Cascading Spanning Tree

In the following definitions we suppose that $G = (V, E)$ is an unempty undirected connected graph. V is the set of its vertices and E is the set of its edges. Let v, w be two vertices of G . We define the **distance between v and w** as the length of the shortest path between v and w and denote it by the symbol $\text{dist}(v, w)$. (The length of a path is equal to the number of edges in the path.) For a nonnegative integer r we denote by symbol $D(v, r)$ the set of all such vertices of V which distance from v is equal to r .

Let H be a subgraph of G and let W be the set of all vertices of H . We say that H is **induced by set W** if, for any pair of vertices $v, w \in W$, $\{v, w\}$ is an edge of H if and only if $\{v, w\}$ is an edge of G . We denote by symbol $\text{INDU}(G, W)$ the subgraph of G induced by set W . Note that graph $\text{INDU}(G, W)$ is defined unambiguously.

Let x be a vertex of G . We say that two vertices $v, w \in V$ are **x -distance bounded** if $\text{dist}(x, v) = \text{dist}(x, w)$ and moreover v and w belong to the same connected component of the graph $\text{INDU}(G, D(x, \text{dist}(x, v)))$.

The relation "vertices v and w are x -distance bounded" is an equivalence relation on V and it thus defines decomposition of V into equivalence classes. Let us denote these classes by symbols $C_0(x), C_1(x), C_2(x), \dots, C_p(x)$, $p > 0$. We reserve symbol $C_0(x)$ for the set $\{x\}$ which itself is an equivalence class in this relation. Note that except for the set $C_0(x)$ the index j of the set $C_j(x)$ is not related to the distance of the vertices of $C_j(x)$ from vertex x .

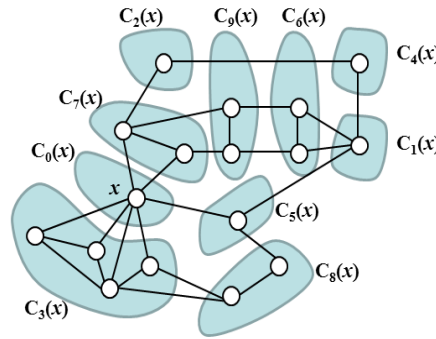


Image 1. The image shows an example of graph with denoted vertex x , and decomposition of the set of its vertices into classes $C_0(x), C_1(x), C_2(x), \dots, C_9(x)$. The set of vertices of induced subgraph $\text{INDU}(G, D(x, 1))$ is the union of classes $C_3(x), C_5(x)$, and $C_7(x)$. The set of vertices of induced subgraph $\text{INDU}(G, D(x, 2))$ is the union of classes $C_1(x), C_2(x), C_8(x)$, and $C_9(x)$. The set of vertices of induced subgraph $\text{INDU}(G, D(x, 3))$ is the union of classes $C_4(x)$ and $C_6(x)$.

Let T be a spanning tree of G , let x be a vertex of G and let $C_0(x), C_1(x), C_2(x), \dots, C_p(x)$ be the decomposition of V into classes defined by relation "vertices v and w are x -distance bounded". We say that T **respects decomposition of V into classes $C_0(x), C_1(x), C_2(x), \dots, C_p(x)$** if for each $j = 1, 2, \dots, p$ holds that T contains exactly one edge $\{v_j, w_j\} \in E(T)$ such that $(v_j \in C_j(x))$ and $(w_j \notin C_j(x))$ and $(\text{dist}(x, v_j) = \text{dist}(x, w_j) + 1)$.

Let W be a set of vertices of G and let T be a spanning tree of G . We say that T **simply spans W** if the graph $\text{INDU}(T, W)$ is connected.

We say that spanning tree T of G is **cascading** if there exists vertex $x \in V$ such that T respects decomposition of G into classes $C_0(x), C_1(x), C_2(x), \dots, C_p(x)$ and T simply spans each class $C_j(x)$, $j = 0, 1, 2, \dots, p$. We say that x defined in the previous sentence is **cascading center of T** . We say that a spanning tree T of G is **minimum cascading** if T is cascading and its weight is minimal among all cascading spanning trees of G .

The task

We are given an undirected integer-weighted graph $G = (V, E, \varphi)$, $\varphi: E \rightarrow \mathbb{Z}$. We have to find the weight of minimum cascading spanning tree of G . The weight of a tree is a sum of weights of all its edges.

Input

The first line of input contains two positive integers N, M separated by space. N represents the order of the graph (number of vertices) and M represents the size of the graph (number of edges). We suppose that graph vertices are labeled $1, 2, 3, \dots, N$. Next, there are M lines of input, each line specifies one edge. The edge specification consists of three integer values separated by spaces. The first two values represent the labels of the end vertices of the edge, the third value represents the weight of the edge. Note that edge weights might be negative. The edges and its end vertices are given in no specific order.

It holds, $2 \leq N \leq 500$, $N-1 \leq M \leq 10000$. The absolute value of any edge weight does not exceed 10^9 .

Output

The output contains a single text line with integer representing the weight of minimum cascading spanning tree of the input graph.

Example 1

Input

```
6 9
1 2 1
1 3 9
1 4 8
1 6 6
2 3 2
3 4 3
```

4 5 4
4 6 7
5 6 5

Output

17

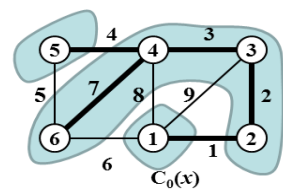


Image 2. The input graph and the solution of Example 1, the cascading center of the minimum cascading spanning tree is vertex 1. Vertex equivalence classes are highlight

[Example2](#)

Input

5 6
1 2 10
1 3 1
2 4 5
3 4 10
3 5 1
4 5 5

Output

12

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout a stderr is available to him/her.

[Public data](#)